

Does chain-of-thought still protect against J-space ablation when problems get hard?

CS 2881 Homework 0 — August 1, 2026 — code: github.com/pgrindehollevik-harvard/cs2881

1 Hypothesis (pre-registered before any experiment; tag prereg-v1)

Gurnee et al. (2026) report that ablating a model’s J-space — zeroing the residual stream’s projection onto the $k=10$ most-active Jacobian-lens directions at every position across a mid-layer band — collapses internal multi-step reasoning while sparing shallow capabilities, and that GSM8K solved *with* chain of thought is far more robust to this ablation than the same problems answered directly. We tested whether that protection persists as problems get harder, on Qwen/Qwen3-4B. **H1** (replication): on GSM8K the random-differenced protection $P_D = [R(\text{CoT}, J) - R(\text{dir}, J)] - [R(\text{CoT}, \text{rand}) - R(\text{dir}, \text{rand})]$ is positive (predicted magnitude $\geq +0.15$), where R = accuracy retention (ablated/clean). **H2** (extension): protection *erodes* with difficulty (GSM8K $\rightarrow$ MATH-500 $\rightarrow$ AIME; within MATH-500, levels 1 $\rightarrow$ 5), because externalization is bandwidth-limited: each step of a harder problem itself needs more un-externalized workspace. Falsification criteria, floor rules, sample sizes, and a hypothesis-blind calibration gate were frozen in docs/PREREGISTRATION.md before any run; all deviations are logged in docs/DEVIATIONS.md.

2 Experiment design

Two prompting modes (CoT vs. direct-answer, both non-thinking mode, matched sampling) $\times$ three conditions (clean / J-space ablation / *per-slot norm-matched* random-direction ablation) $\times$ three datasets (GSM8K $n=150$; MATH-500 stratified 30/level; AIME 2024+2025, all 60). The random control removes an exactly equal-norm component along random directions at the same (position, layer), so anything surviving the P_D subtraction is J-space-specific rather than generic damage. Before the main grid may run, a hypothesis-blind stop-gate requires: a significant positive-control drop under J-ablation (exact McNemar), $\geq 2\times$ the matched-random drop, output degeneration $< 10\%$, and ordinary next-token prediction on held-out wikitext surviving (> 0.80 top-1 agreement with clean; our pre-registered quantification of the paper’s selectivity finding, whose exact values appear only in its Fig. 22 graphic). Strength (layer band, k) is calibrated *only* on these criteria, never on CoT-vs-direct gaps.

3 Experimental details

No ablation code exists in the companion repo; we implemented it as forward hooks: per (position, band-layer), readout logits = $\text{lm_head}(\text{rmsnorm}(J_\ell h))$, top-10 token ids (recomputed per position and layer, ids in the clean model’s top-10 next-token predictions exempted via a parallel clean stream over the same tokens), directions $v_t = J_\ell^\top (\gamma \odot W_U[t])$, joint projection removed by regularized batched solve. Ablation applies to prompt and every generated token. Pre-fitted lens: neuronpedia/jacobian-lens (fit on this exact checkpoint); it beats a logit-lens baseline at ranking silent multi-hop intermediates (pass@5 0.41 vs 0.37; pass@1 0.20 vs 0.19) though not on arithmetic order-of-operations readouts at rank 1 — the pre-registered gate (win on either eval) passed before use. Percent-depth transfer of the paper’s Sonnet band gives candidates {14–24, 14–31} of 36 blocks (light = 14–19). Compute: one M4 Pro (48 GB, MPS); ablated decode 3.4–7.2 tok/s across calibration rungs, median ≈ 6 (dual streams + per-layer readouts). Runs are chunk-resumable (64-token persistence): macOS update preparation repeatedly SIGKILLED the process (90 s lifetimes at the worst), which cost wall-clock but, by construction, no data. Grading: math-verify on the last `\boxed{\}`; artifact classes (no-answer / truncation / degeneration / wrong-answer) tracked separately. 7 unit tests certify the intervention (identity when disabled, projections verifiably removed, exemption honored, per-slot norm match, seeded determinism, chunked = unchunked).

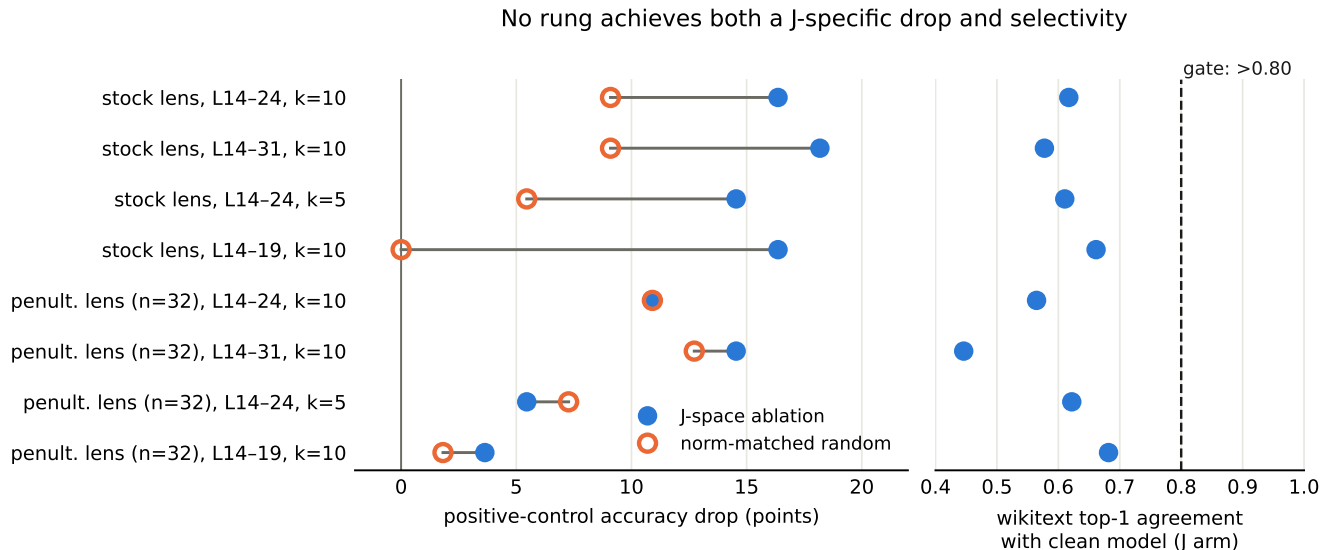
4 Experimental results

Positive-control calibration. The paper’s control (“multihop”: silent two-hop factual composition, e.g. legs of the animal that spins webs) proved unusable — clean Qwen3-4B scores only 0.42, itself a small-model datum — so the pre-registered fallback control (“order-ops”: arithmetic order-of-operations completions, clean 0.745) was used. All four pre-registered rungs *failed* the stop-gate, so the main grid did not run on the off-the-shelf lens:

Rung	J drop (pts)	random drop	wikitext top-1 (J / rand)	gate
14–24, $k=10$	−16.4 ($p=.004$)	−9.1	0.62 / 0.79	fail
14–31, $k=10$	−18.2	−9.1	0.58 / 0.73	fail
14–24, $k=5$	−14.5	−5.5	0.61 / 0.82	fail
14–19, $k=10$ (light)	−16.4 ($p=.012$)	0.0	0.66 / 0.84	fail

The J-directions are causally real on Qwen3-4B — in the light band, norm-matched random damage costs *nothing* while J-ablation costs 16 points ($n=55$ paired items, exact McNemar $p=.012$; degeneration 3%) — but they are

not *selective*: 34–42% of ordinary next-token predictions change, where the paper finds ablation leaves “the bulk of ordinary text prediction intact” (§3.5.2). Per the pre-registered contingency (the stock lens targets the final layer; the paper’s recipe targets the penultimate layer to avoid last-block artifacts), we refit a penultimate-target lens ($n=32$ prompts; 17 h of MPS backward passes made the registered $n\approx 100$ infeasible — deviation 4, with its interpretation constraint committed *before* the re-run) and re-ran the ladder. All four rungs failed again, in a diagnostic pattern: the refit lens’s “J-directions” behaved *identically to the random control* (drops 10.9 vs 10.9; 14.5 vs 12.7), while selectivity worsened at the $k=10$ rungs (0.45–0.56) and stayed comparable at the gentler rungs (0.62–0.68) — nowhere near the 0.80 gate. This pattern is consistent with estimation noise dominating the small fit, which would corroborate the stock lens’s J-vs-random differential as real structure; deviation 4’s registered confound means we treat this reading as suggestive rather than established.



5 Analysis of results

Verdict. H1/H2 are unanswerable as posed on this model: the pre-registered precondition for interpreting a CoT-vs-direct comparison — an ablation that is J-specific *and* spares ordinary prediction — was not achievable at any of eight settings across two lenses. We report the negative replication rather than an uninterpretable grid. **What is positively established:** (i) the Jacobian lens transfers to Qwen3-4B as a *readout* (it beats the logit lens at surfacing silent multi-hop intermediates); (ii) its directions are *causally load-bearing* — the light-band result (16-point J drop, zero norm-matched-random drop, outputs coherent) is a clean causal dissociation; (iii) what fails to transfer is the paper’s *selectivity* signature. **Two readings remain.** (a) Small models route ordinary next-token prediction *through* verbalizable representations — the workspace/automatic separation is an emergent property of scale (consistent with the paper’s Haiku 4.5 warning, and with our finding that clean Qwen3-4B scores only 0.42 on two-hop silent composition); (b) lens estimation quality at 4B (the stock lens early-stopped at $n=479$; final-layer target) smears workspace directions with general-prediction components. Our refit could not adjudicate (b) at a feasible fit size — reported as unresolved, not refuted. **Next with more compute:** the same pre-registered grid on Qwen3-14B/32B (pre-fitted lenses exist), where reading (a) predicts the gate passes and (b) predicts it keeps failing; a full-budget penultimate lens; and thinking-mode traces. All code, prompts, seeds, the full 8-rung calibration records (including per-item outcomes), and the crash-tolerant pipeline are in the repository for exact reuse; the main grid’s records intentionally do not exist.